

BUSINESS REQUIREMENTS DOCUMENT (BRD)

AI-Powered CMO Response Time Reduction & Incident Management Automation System

Defines the business needs, scope, benefits and high-level requirements for the incident-management automation initiative.

Document Owner	Status	Version	Classification
Business Analyst / Project Lead	Portfolio-ready	1.0	Sanitized public case study

1. Purpose

This Business Requirements Document defines the business need for an AI-powered system that improves telecom incident response, CMO management and operational visibility. It describes what the business requires; the detailed system behavior is defined in the Functional Requirements Document (FRD).

2. Business Context and Problem Statement

Incident notifications were manually monitored from operational communication channels. The process relied on people to read raw alerts, determine criticality, identify ownership, follow up on open incidents and compile reports. This created response delays, inconsistent escalation and limited leadership visibility.

- Critical alerts could take approximately 10 to 45 minutes to reach the relevant response team.
- Follow-ups and escalation depended on manual effort and individual availability.
- Critical fiber incidents were monitored through periodic data extraction and filtering.
- Manual reporting reduced time available for proactive incident management.

3. Business Objectives

- Reduce CMO-related incident dockets and improve response effectiveness.
- Provide near real-time operational visibility for active and critical incidents.
- Standardize alert routing, ownership, follow-ups and escalation.
- Reduce manual monitoring and reporting effort.
- Strengthen SLA governance, accountability and leadership decision support.

4. Stakeholders

Stakeholder	Role	Business Need
Operations Leadership	Executive sponsor	KPI visibility and improved operational performance
Fiber Operations Team	Primary users	Timely, structured alerts and incident tracking

Stakeholder	Role	Business Need
NOC Team	Incident source	Reliable transfer of incident information
Area Managers	Response owners	Clear ownership, reminders and escalation
Circle Operations Head	Escalation authority	Visibility of critical exceptions and trends
Project Lead	Requirement and delivery owner	Documented scope, validation and successful adoption

5. Current State (AS-IS)

A raw incident notification is received, manually read and interpreted, then shared with the relevant team. The team follows up through manual messages and escalates based on individual judgement. Open-incident tracking and management reporting are created separately, usually after the event.

6. Future State (TO-BE)

The system receives approved incident updates, applies defined business rules to identify relevant alerts, produces structured notifications, routes them to owners, triggers reminders and escalations, and makes current status and KPIs available in an operational dashboard. Human owners retain responsibility for diagnosis, resolution and authorized decisions.

7. Scope

7.1 In Scope

Module	In-Scope Business Capability
Reactive Incident Automation	Alert intake, intelligent filtering, criticality identification, ownership tagging, structured notification, reminders, escalation, restoration notification and lifecycle tracking.
Proactive Incident Monitoring	Critical incident capture, area-wise pending visibility, reminder generation, KPI reporting and executive dashboard publication.

7.2 Out of Scope

- Network fault resolution, field maintenance and hardware replacement.
- Core network configuration, capacity planning and infrastructure upgrades.
- Replacement of existing NOC systems or communication platforms.
- Automated decisions that require authorized human approval.

8. Business Requirements

ID	Business Requirement	Priority	Acceptance Measure
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ID	Business Requirement	Priority	Acceptance Measure
BR-01	The business requires incident notifications to be captured from approved operational channels.	High	Relevant alerts are available for processing.
BR-02	The business requires critical and service-impacting incidents to be identified promptly.	High	Defined alert types are correctly flagged.
BR-03	The business requires a structured alert to be sent to the relevant response owner.	High	Correct owner receives the structured notification.
BR-04	The business requires automated follow-ups for open incidents.	High	Reminder is issued at the defined interval until closure.
BR-05	The business requires escalation of long-running incidents according to approved rules.	High	Escalation is sent when the threshold is met.
BR-06	The business requires restoration updates to be communicated to stakeholders.	High	Closure notification is issued for a valid restoration update.
BR-07	The business requires proactive visibility of critical open incidents by area.	High	Dashboard displays current pending incident status.
BR-08	The business requires operational KPI reporting for leadership.	Medium	Dashboard shows agreed CMO and incident metrics.
BR-09	The business requires an auditable record of automation activity and exceptions.	Medium	Logs capture processing status, failures and actions.
BR-10	The business requires confidential identifiers and credentials to be protected.	High	No sensitive information appears in public artefacts or logs.
BR-11	The business requires a manual fallback process during automation or source-system disruption.	Medium	Users can follow the approved fallback process.
BR-12	The business requires the automation to be maintainable when input formats change.	Medium	Rules can be reviewed and updated after validation.

9. Business Rules

- Only approved operational sources and alert categories are eligible for automated processing.
- An incident is routed only when the required identifying data is available; otherwise it is logged for manual review.
- Duplicate alerts within the defined matching window must not create duplicate notifications.
- Escalation follows approved duration and severity thresholds; the system does not override human authority.
- Restoration communication is sent only after a valid restoration update is recognized.
- Public portfolio documentation must use sanitized names, IDs and sample data.

10. Success Criteria and KPIs

KPI	Baseline	Outcome / Target
Total CMO-related incident dockets	607	494; 19% reduction
30-60 minute dockets	71	58; 18% reduction
1-2 hour dockets	161	104; 35% reduction
Dockets open for more than 3 hours	248	202; 19% reduction
Incident visibility	Manual / delayed	Near real-time dashboard visibility
Follow-up tracking	Manual	Automated and standardized

11. Assumptions, Constraints and Risks

Type	Detail	Response
Assumption	Approved operational communication channels continue to provide usable incident updates.	Validate source availability during implementation and operations.
Constraint	System performance depends on source format, platform availability and data quality.	Use monitoring, logging and an approved manual fallback.
Risk	Incoming message formats may change and reduce classification accuracy.	Maintain configurable rules and periodic validation.
Risk	Users may continue manual practices rather than adopt structured alerts.	Provide simple job aids, onboarding and feedback-led refinement.

12. Approval

This BRD is a sanitized portfolio case study. It is intended to demonstrate business analysis practices and does not disclose confidential organizational information.

Approver Role	Name / Signature	Date
Project Sponsor / Operations Leadership	_____	_____
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Project Lead	_____	_____
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Primary Business Representative	_____	_____
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